

Threat Hunting with Splunk

Threat Investigation of Botsv1 dataset in Splunk

Aayush Dhakal

Bachelors in Computer Networking and IT Security

Islington College, Nepal

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1 Dataset Overview

1. Dataset: Boss of the Splunk version 1 (Botsv1)
2. Target Network: Wayne Enterprises (hybrid cloud/on-premise enterprise infrastructure).
3. Data Ingested: Windows Event Logs, Sysmon, IIS Web Server Logs, Network Streams (Stream:HTTP, Stream:DNS), Fortinet Firewall logs, and Suricata IDS alerts.
4. Web server IP: 192.168.250.70
5. Attack: Defacement of the website
6. Date: 2016

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2 Hunting

2.1 Reconnaissance

In order to build the SPL that can be used to narrow down the huge logs of the dataset, certain information is needed.

1. The dataset used is botsv1.
2. The Wayne Enterprise has a web server.
3. The attackers use automated tools to enumerate websites which can create a lot of 404 errors. A specific IP address that causes a lot of 404 errors is likely the attacker.

So, with this information, the following SPL was built:

index=botsv1 sourcetype=stream:http status=404 | stats count by src_ip

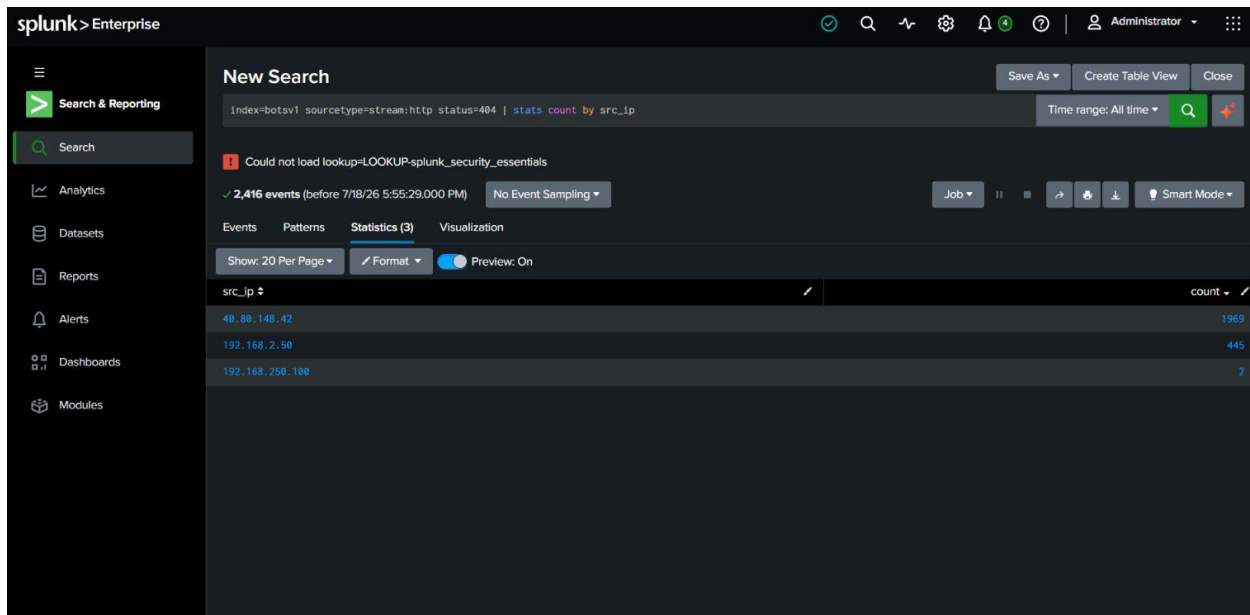


Figure 1 Query 1 result

The IP address 40.80.148.42 has a very high count of 1969. This means, this IP address is likely using automated vulnerability scanning tool and causing 404 errors.

The following query was used to inspect the header and find the tool used by the attacker:

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index=botsv1 sourcetype=stream:http src_ip="40.80.148.42" | stats count by src_headers

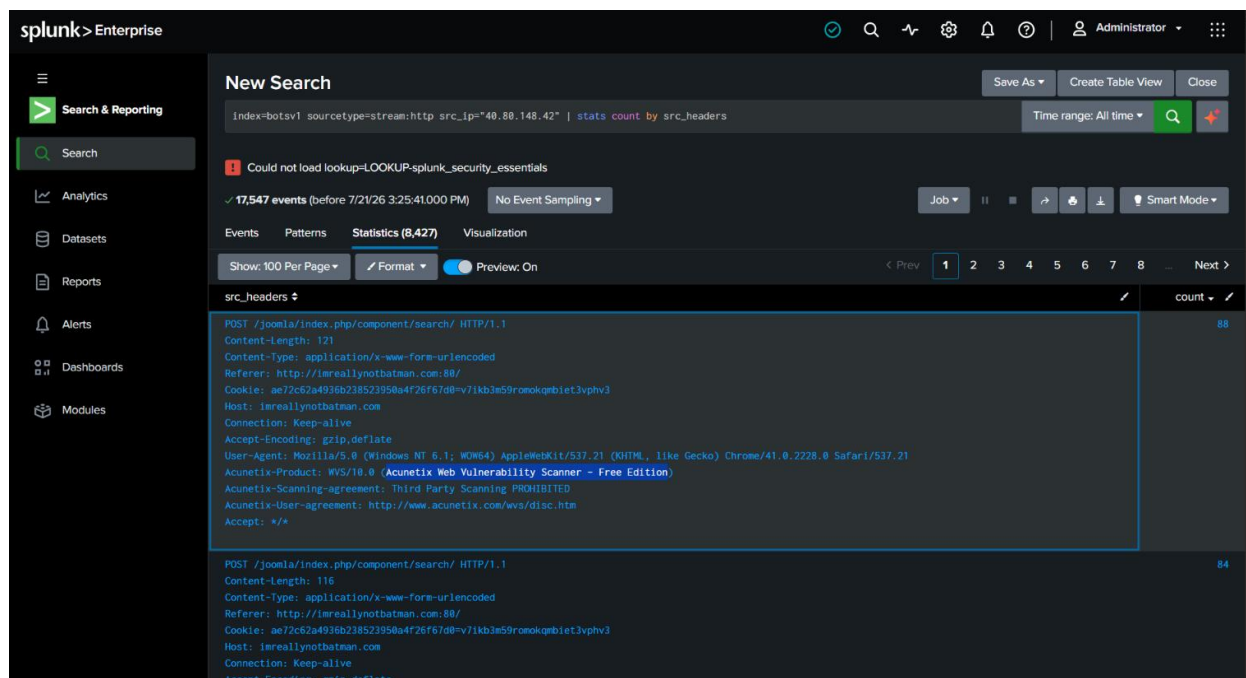


Figure 2 Query II result

The attacker used free edition of Acunetix Web Vulnerability Scanner.

The next step is to identify if the attacker successfully loaded the website. The following query can be used to filter this:

index=botsv1 sourcetype=stream:http src_ip="40.80.148.42" status=200 | table _time, http_method, uri, status

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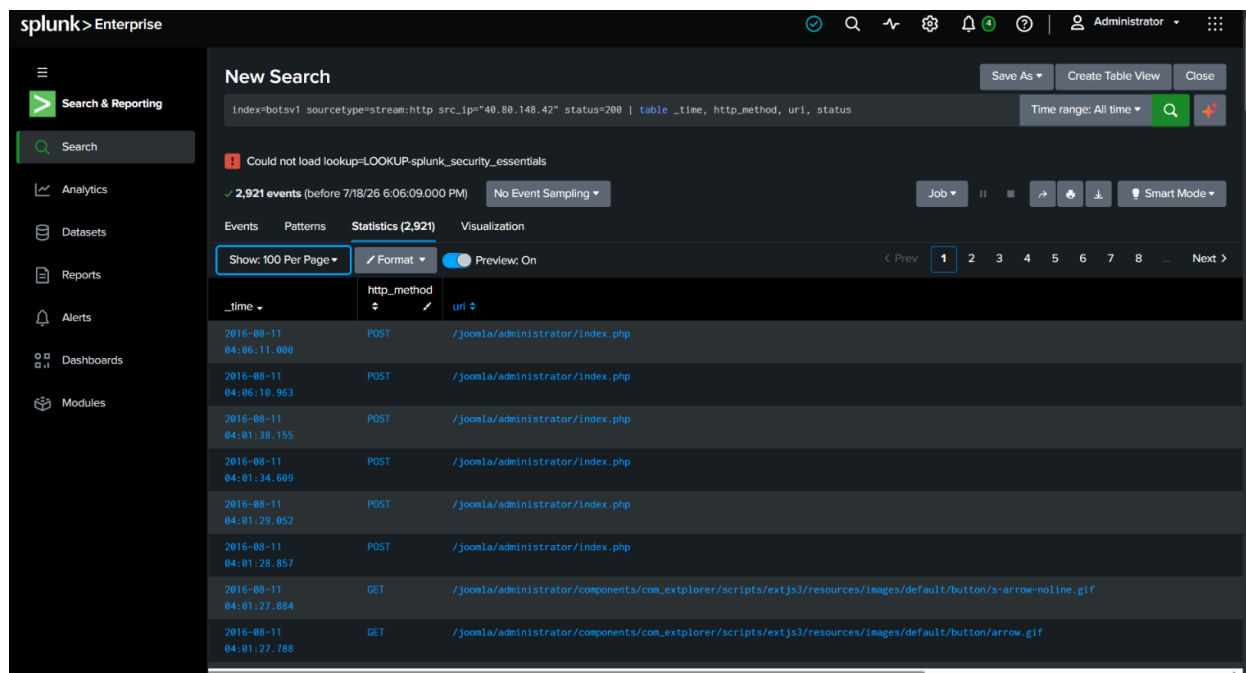


Figure 3 Query III result

The result shows that the attacker was successful in various searches in Joomla/administrator/index.php. Joomla is a popular Content Management System (CMS). Sorting the time field can help identify the chronological steps of the attacker.

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2.2 Delivery

The attacker was trying to brute-force the way into the server targeting Joomla component pages. The attacker may have used brute-force attack to login to the administrator page. Following query can be used to investigate this:

```
index=botsv1 sourcetype=stream:http src_ip="40.80.148.42" uri="/joomla/administrator/index.php" http_method=POST | table _time, form_data
```

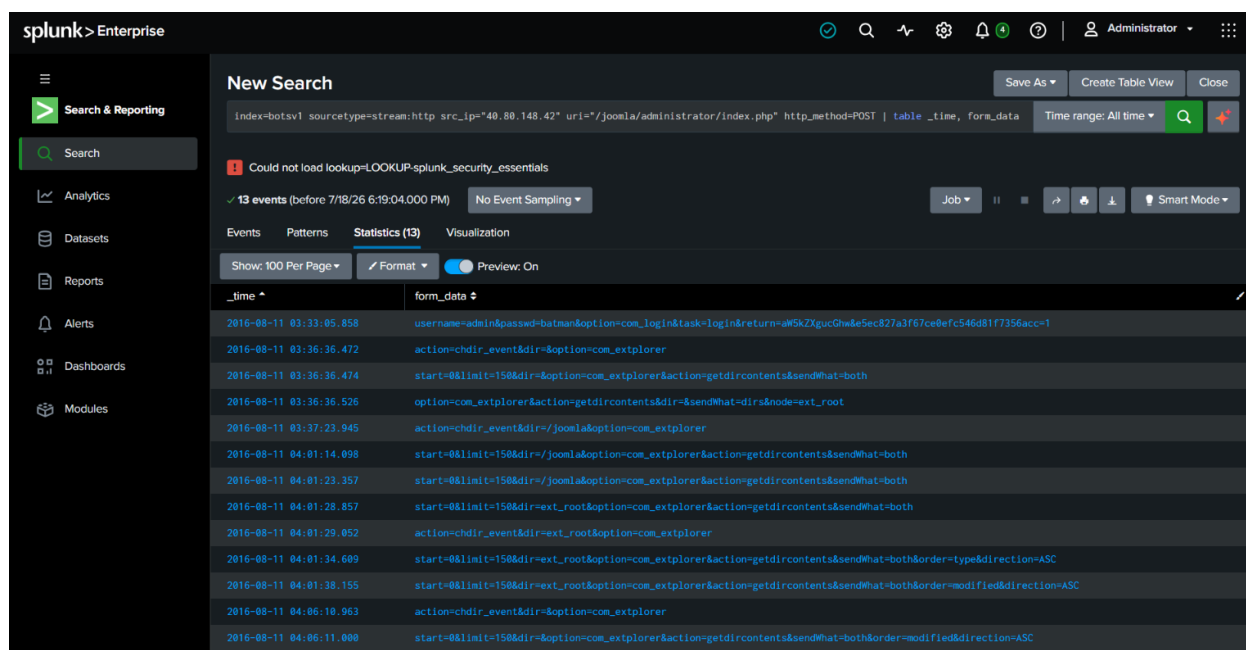


Figure 4 Query IV result

The first log shows that the attacker had tried to log in with username admin and password batman. Investigating that event gives some information such as the site name was iamreallynotbatman.com and the server IP address was 192.168.250.70. Also, the attacker only tried one credential and succeeded. The attacker may have used a different IP address to brute-force.

The following query was used to identify what IP addresses were causing the brute-force:

```
index=botsv1 sourcetype=stream:http uri="/joomla/administrator/index.php" http_method=POST form_data=*passwd* | rex field=form_data "passwd=(?<password>[^\&]+)" | stats count by src_ip
```

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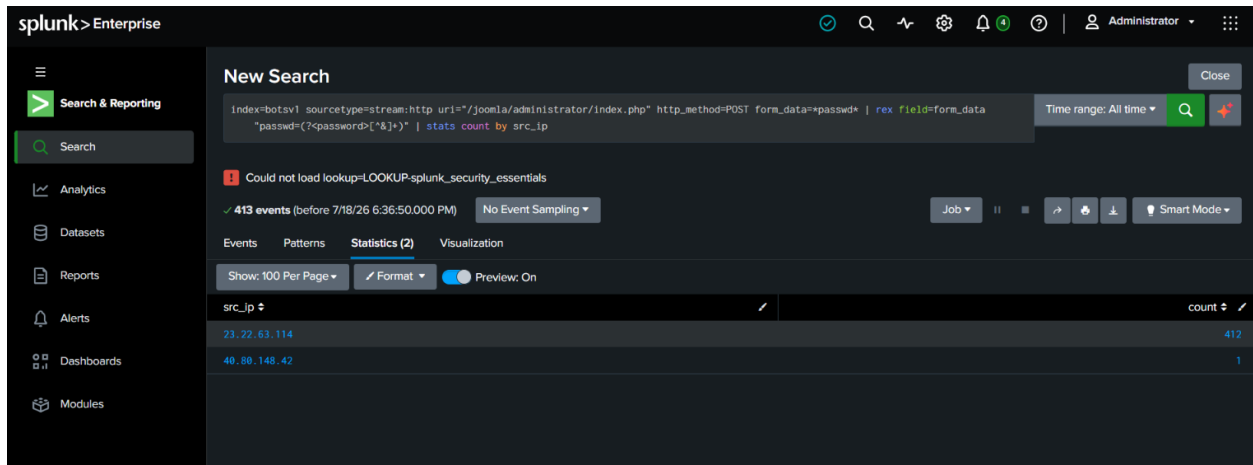


Figure 5 Query V result

The result displayed that 412 attempts were made by the IP address 23.22.63.114.

The following query gives the result of all the username and password the IP address tried to brute-force:

```
index=botsv1          sourcetype=stream:http          uri=\"/joomla/administrator/index.php\"  
http_method=POST form_data=*passwd* src_ip=\"23.22.63.114\" | table _time, form_data
```

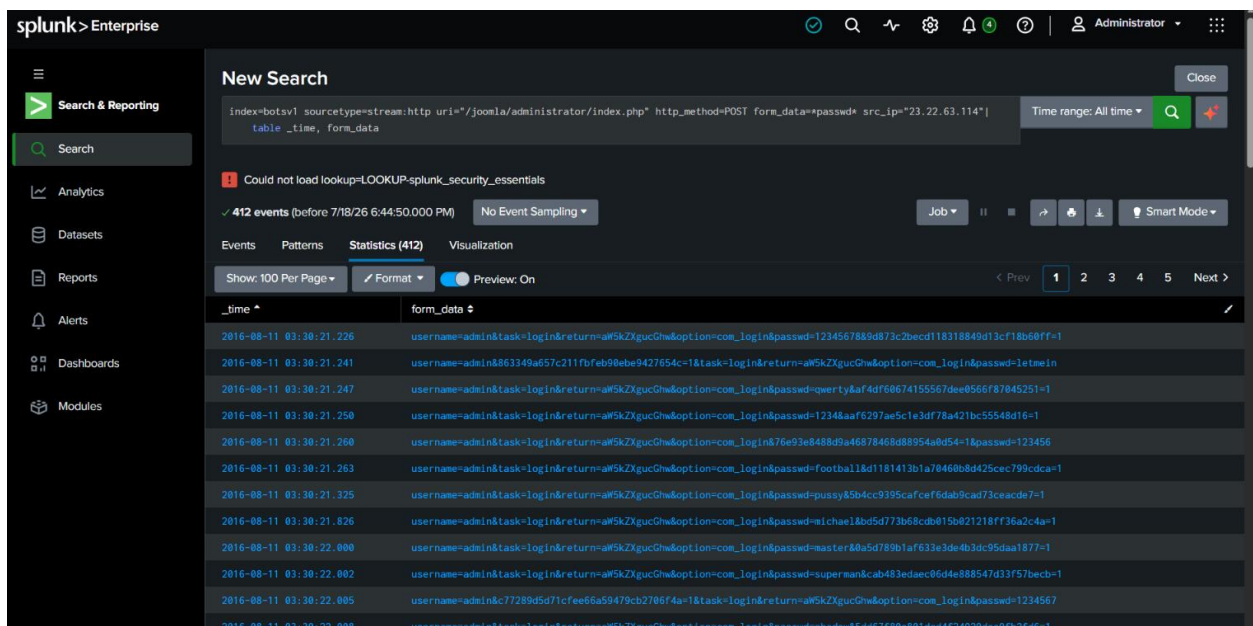


Figure 6 Query VI result

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The correct credential (username=admin and password=batman) has already been identified. The following query is used to see the event when the brute-force attempt from 23.22.63.114 matched this credential.

```
index=botsv1 sourcetype=stream:http uri="/joomla/administrator/index.php" http_method=POST form_data="*passwd=batman"* src_ip=23.22.63.114 | table _time, form_data
```

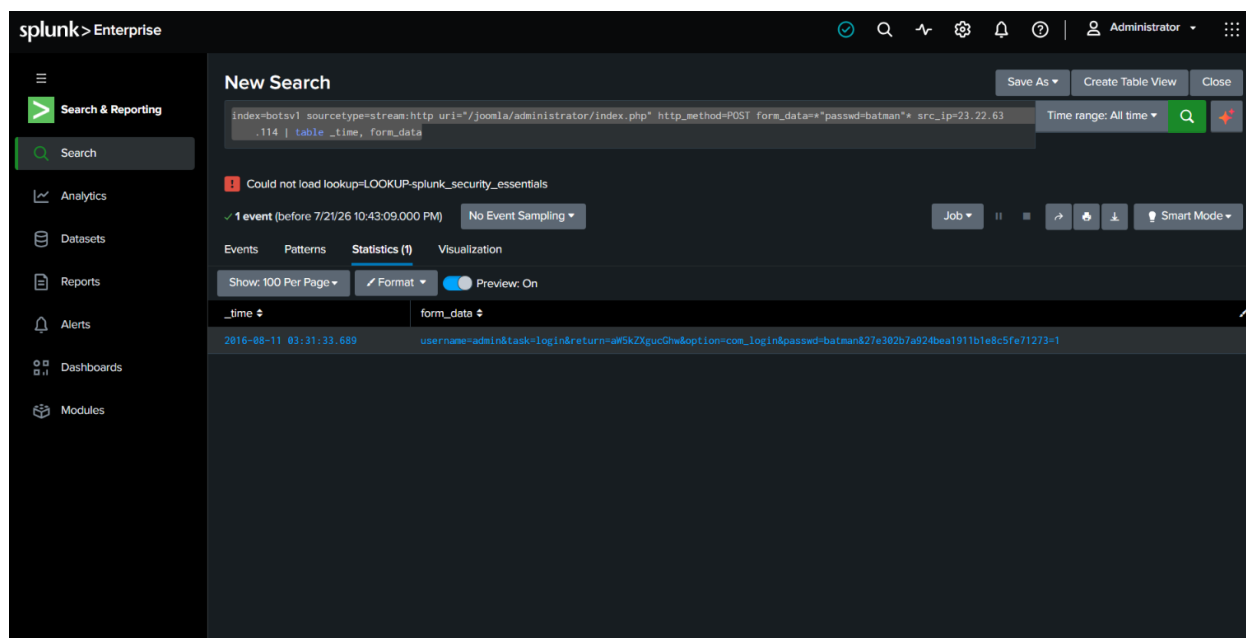


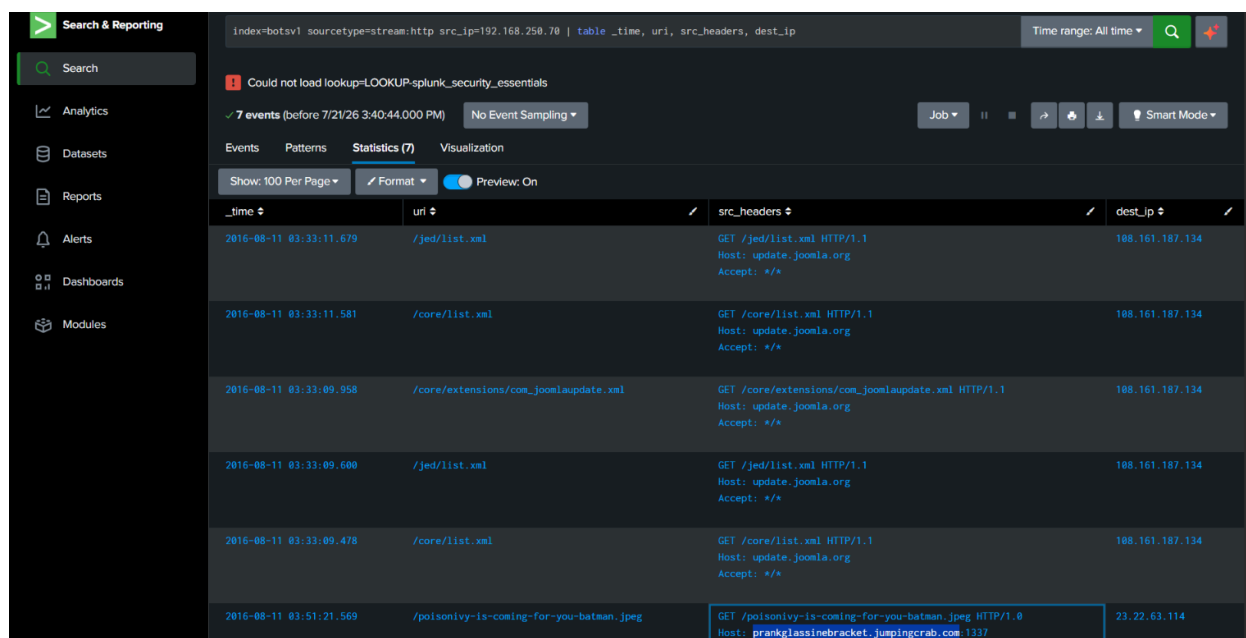
Figure 7 Query VII result

So, the correct credential was identified in 3:31:33 and the attacker logged on at 3:33:05.

The next step is to identify the domain name of the IP address. Following query was used to achieve this:

```
index=botsv1 sourcetype=stream:http src_ip=192.168.250.70 | table _time, uri, src_headers, dest_ip
```

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Search & Reporting

index=botsv1 sourcetype=stream:http src_ip=192.168.250.70 | table _time, uri, src_headers, dest_ip

Time range: All time

Could not load lookup=LOOKUP-splunk_security_essentials

7 events (before 7/21/26 3:40:44.000 PM) No Event Sampling

Job

Events Patterns Statistics (7) Visualization

Show: 100 Per Page Format Preview: On

_time	uri	src_headers	dest_ip
2016-08-11 03:33:11.679	/jed/list.xml	GET /jed/list.xml HTTP/1.1 Host: update.joomla.org Accept: */*	108.161.187.134
2016-08-11 03:33:11.581	/core/list.xml	GET /core/list.xml HTTP/1.1 Host: update.joomla.org Accept: */*	108.161.187.134
2016-08-11 03:33:09.958	/core/extensions/com_joomlaupdate.xml	GET /core/extensions/com_joomlaupdate.xml HTTP/1.1 Host: update.joomla.org Accept: */*	108.161.187.134
2016-08-11 03:33:09.600	/jed/list.xml	GET /jed/list.xml HTTP/1.1 Host: update.joomla.org Accept: */*	108.161.187.134
2016-08-11 03:33:09.478	/core/list.xml	GET /core/list.xml HTTP/1.1 Host: update.joomla.org Accept: */*	108.161.187.134
2016-08-11 03:51:21.569	/poisonivy-is-coming-for-you-batman.jpeg	GET /poisonivy-is-coming-for-you-batman.jpeg HTTP/1.0 Host: pranglassinebracket.jumpingcrab.com:1337	23.22.63.114

Figure 8 Query VIII result

The attacker uses the server with the domain name pranglassinebracket.jumpingcrab.com to upload malware or deface the website.

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2.3 Exploitation

The attacker had accessed the administrator privilege via credentials. So, they would exploit the CMS or the web application.

1. com_extplorer is a file manager in Joomla. The attacker could play around this.
2. The attacker could execute commands or upload malicious files.
3. The attacker could try to perform XSS, SQL injection and other payload-based attacks.
4. The attacker could throw a lot of payloads resulting in spike of 500 errors.

The brute-force attempt occurred around 03:30 on 2016-08-11. So, the exploitation would occur shortly after this time period.

So, the following queries were used to test these:

```
index=botsv1 sourcetype="stream:http" http_method=POST "com_extplorer"  
src_ip="40.80.148.42" | table _time, uri, form_data
```

_time	uri	form_data
2016-08-11 03:37:23.945	/joomla/administrator/index.php	action=chdir_event&dir=/joomla&option=com_extplorer
2016-08-11 03:36:36.472	/joomla/administrator/index.php	action=chdir_event&dir=/joomla&option=com_extplorer
2016-08-11 03:36:36.526	/joomla/administrator/index.php	option=com_extplorer&action=getdircontents&dir=/joomla&node=ext_root
2016-08-11 03:36:36.474	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both
2016-08-11 04:01:29.052	/joomla/administrator/index.php	action=chdir_event&dir=/joomla&option=com_extplorer
2016-08-11 04:01:23.357	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both
2016-08-11 04:01:14.098	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both
2016-08-11 04:06:11.000	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both&order=modified&direction=ASC
2016-08-11 04:06:10.963	/joomla/administrator/index.php	action=chdir_event&dir=/joomla&option=com_extplorer
2016-08-11 04:01:28.857	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both
2016-08-11 04:01:38.155	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both&order=modified&direction=ASC
2016-08-11 04:01:34.609	/joomla/administrator/index.php	start=0&limit=150&dir=/joomla&option=com_extplorer&action=getdircontents&sendWhat=both&order=type&direction=ASC

Figure 9 Query IX result

The attacker just enumerated the com_extplorer.

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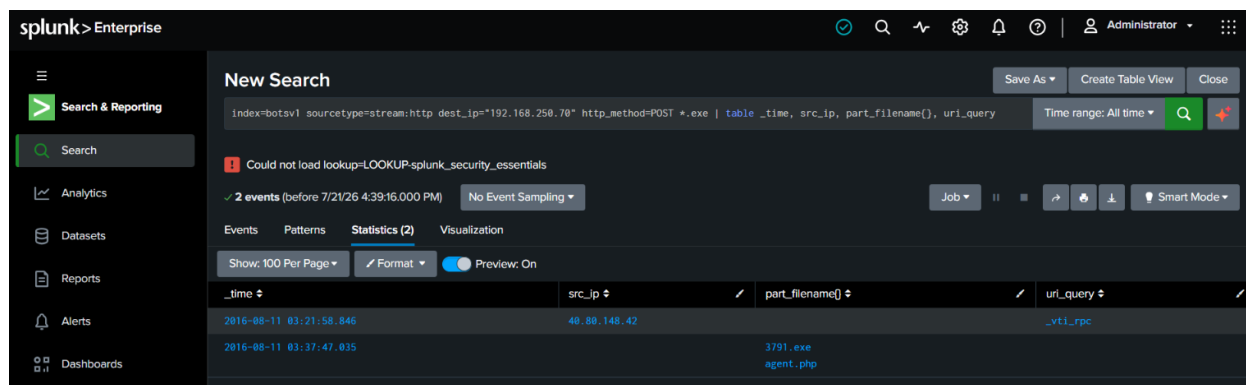
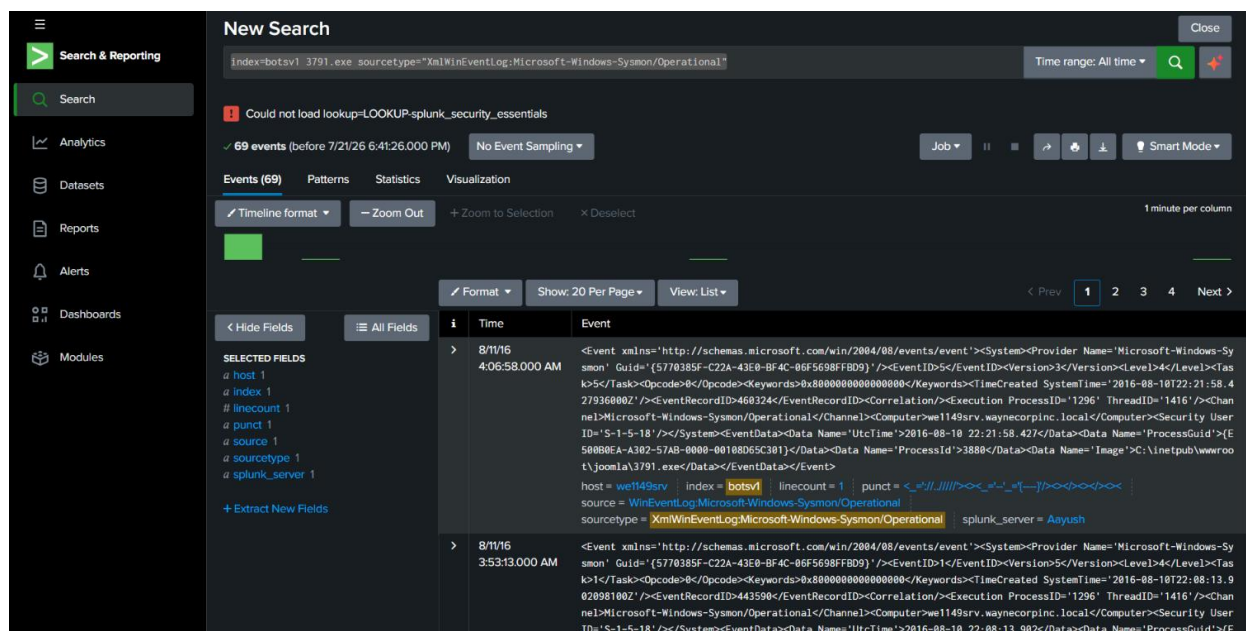


Figure 11 Query XI result

There were two files: 3791.exe and agent.php.

In order to confirm that the file was executed, the following query was used:

`index=botsv1 3791.exe sourcetype="XmlWinEventLog:Microsoft-Windows-Sysmon/Operational"`



[illegible]

The resulting log shows the following information:

- EventID: 1: This confirms a new process was started on the system.
- Image: C:\Windows\SysWOW64\cmd.exe: This is the process that was just created.
- CurrentDirectory: C:\inetpub\wwwroot\joomla\ : The command prompt was launched inside the Joomla web application folder.
- ParentImage: C:\inetpub\wwwroot\joomla\3791.exe: This process started cmd.exe.

There are also hashes of the program execution.

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2.5 Action on objectives

The attacker defaces the website. In order to see the defacement, following query was used:

index=botsv1 sourcetype="stream:http" src_ip=192.168.250.70 | stats count by uri

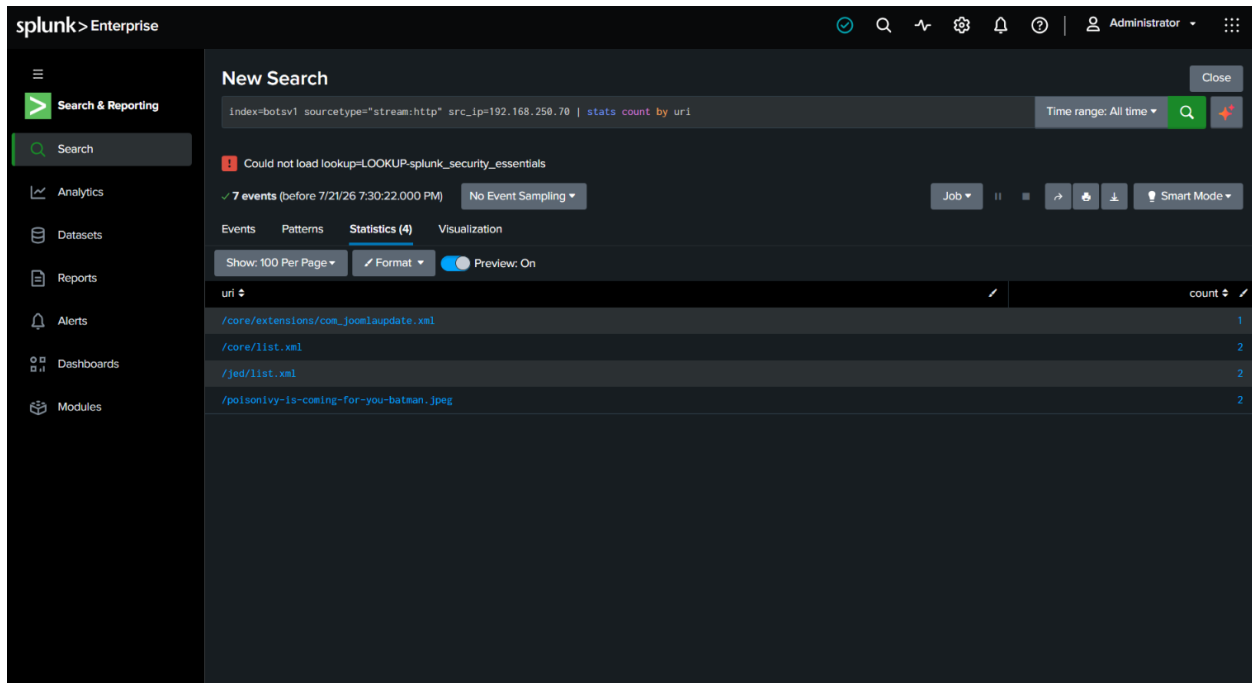


Figure 13 Query XIII result

There is an image `poisonivy-is-coming-for-you-batman.jpeg` in the result. The attacker had defaced the website with this image.

3 Findings

1. The attacker (40.80.148.42) used free edition of Acunetix Web Vulnerability Scanner to scan vulnerabilities in the website iamreallynotbatman.com (192.168.250.70). The website was also using the CMS Joomla.
2. The attacker tried to brute-force using the IP address 23.22.63.114.
3. The attacker made first brute-force attempt in 3:30:21 and successfully logged on with the correct credentials in 3:33:05. So, the attacker was successful within 3 minutes.
4. The attacker gained admin privilege by getting credentials via brute-force attack.
5. The attacker uses the server with the domain name prankglassinebracket.jumpingcrab.com to upload malware or deface the website.
6. The attacker executed 3791.exe.
7. The attacker defaced the website by posting the image poisonivy-is-coming-for-you-batman.jpeg.